OM MULCHANDANI

INTERDISCIPLINARY TECHNOLOGIST

DATA / BIOINFORMATICS / ARTIFICIAL INTELLIGENCE

Email: ommulchandani2108@gmail.com
Location: Rajkot, Gujarat, India
Portfolio: om-mulchandani.tech
Focus: Systems, Data & Computational Biology

// PROFESSIONAL SUMMARY

Interdisciplinary technologist with 2+ years of practical foundation in data analytics, currently pursuing a B.Tech in Bioinformatics and actively developing intelligent computational systems. Passionate about exploring the space between disciplines — where biological inquiries become computational problems, raw telemetry becomes structured workflows, and AI models evolve into useful operational products. Relentlessly curious, systems-oriented, and comfortable executing across unfamiliar technological domains.

// EXPERIENCE & APPLIED TRAJECTORY

TOPS TECHNOLOGIES PVT. LTD. Rajkot, India | Ongoing Training
Professional Data Analytics & BI Track

- Mastering end-to-end data pipelines: transforming unstructured transactional data into normalized relational databases.
- Authoring advanced SQL queries utilizing window functions, subqueries, complex multi-table joins, and stored procedures.
- Designing interactive business intelligence dashboards in Power BI and Excel, utilizing DAX measures and Power Query ETL steps.
- Conducting statistical and exploratory data analysis (EDA) using Python (Pandas, NumPy, Matplotlib).

HELIX ANALYTICS 2024 | Fictional / Demo Project Case
Entry-Level Data Analyst

- Executed data cleaning and validation workflows on messy operational datasets to establish reproducible data hygiene.
- Constructed analytical views and automated recurring executive reporting, reducing manual compile cycles by 60%.
- Collaborated with team leads to define core metrics and translated business criteria into structured SQL queries and dashboards.

AXIOM LABS 2025 | Fictional / Demo Project Case
AI Research Intern

- Engineered prototype Retrieval-Augmented Generation (RAG) pipelines for parsing, chunking, and querying scientific documents.
- Experimented with dense vector embeddings, semantic search reranking, and multi-agent tool execution loops.
- Prototyped automated knowledge extraction pipelines converting raw PDF research texts into structured JSON knowledge nodes.

// SELECTED RESEARCH PROJECTS & EXPERIMENTS

GENOME X — Computational Biology Visualization System 2026
An interactive web-based scientific tool for visualizing genomic loci, mutation maps (CFTR, TP53, BRCA1, APOE), and tissue-specific RNA-seq expression heatmaps. Implemented with dynamic coordinate SVG rendering and biological sequence analysis algorithms.

ATLAS — AI Research Intelligence System 2025
A semantic research engine that ingests technical research papers, constructs citation and concept graphs, and generates grounded extractive synthesis using high-dimensional vector embeddings and cross-encoder rerankers.

PROJECT PULSE — AI × Bioinformatics Research Experiment 2026
An exploratory computational framework investigating how modern AI algorithms detect non-linear patterns, entropy changes, and anomalous biomarker signatures across multimodal biological streaming data.

// EDUCATION

BACHELOR OF TECHNOLOGY (B.TECH) IN BIOINFORMATICS Expected 2026/2027
Undergraduate Degree Program | Ongoing

Core Coursework: Computational Biology, Molecular Genetics, Database Management Systems, Data Structures & Algorithms, Biostatistics, Genomic Sequence Analysis.

// TECHNICAL SKILLS & CAPABILITIES

DATA ANALYTICS:
Advanced Excel (Formulas, Pivot Tables, Lookups), SQL (PostgreSQL, MySQL), Power BI (DAX, Modeling), Data Cleaning, Pandas, NumPy, Exploratory Data Analysis (EDA).

COMPUTATION & CODE:
Python, Vanilla JavaScript (ES6+), RESTful APIs, ETL Pipelines, Process Automation, Linux/CLI, Git & Version Control.

ARTIFICIAL INTELLIGENCE:
LLM Prompt Engineering, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Embeddings, AI Agents, Tool-Calling Architectures.

BIOINFORMATICS:
Computational Biology Algorithms, Genomic Sequence Analysis, Biological Database Querying (NCBI, Ensembl, ClinVar), Scientific Computing.